NETWORK ANALYSERS Moving To Lasers For Increased Effectiveness

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Benchtop network analysers have seen huge design changes in the last few years, mostly to ensure an integrated approach to measurements on active devices such as amplifiers and frequency converters. With the growing popularity of dual-source network analysers, measurements can be done with a single instrument. These are also much faster as the sources and receivers are on the network analysers. But more on that later; let us first look at the changes in network analysers over the past few years.

Requirement influencing design

Vector network analysers (VNAs) have proven themselves to be accurate, owing to their sophisticated test set design and the kind of applications these are used in. Changes in these systems are concentrated on developing innovative calibration algorithms, measurement applications and digital signal processing techniques. These techniques, for example, are applied once the signal is down-converted and digitised by an analogue-to-digital converter.

System design has changed to accommodate advanced measurements like built-in pulse generators and modulators that help characterise radio frequency or microwatt components in pulsed mode. There is a lot of development in new concepts in millimetre range for a range of applications, especially related to RADAR.

With two-source network analysers having a built-in combiner, these measurements can be done very easily. Built-in second source and combiner networks help with accurately measuring and calibrating intermodulation distortion (IMD) for amplifiers and frequency converters.

Special low-noise receivers are being used for source-corrected accurate noise figure measurements for low-noise amplifiers and receivers. True-mode stimulus allows for fast and fully error-corrected mixed-mode S-parameter measurements.

Pulsed measurements. Pulsed measurements for devices such as transmitter and receiver modules, and pulse amplifiers, and noise figure measurements on converters and amplifiers, are some of the recent features being focused on for the design of network analysers. Harmonic specifications of sources improve with the use of filters in more advanced network analysers to enable better mixed measurements and IMD measurements. Third-order IMD in amplifiers requires two sources and a spectrum analyser with traditional approach.

But it is not just the feature sets that are changing. Analysers are going through physical changes, too.

Aiming for a better handheld device. Connectivity with the network analyser is simpler, and measurements over a wide frequency range can be made much faster. Vendors now aim for smaller, lightweight devices. "Smaller the size, easier the usage and lower the power losses," is how one vendor succinctly puts it.

This has led to an increase in field testing equipment as well. Handheld VNAs cover up to 50GHz range, and integrate full two-port VNAs, spectrum analysers, vector voltmeters, power meters and channel scanners, among other measurement devices. Field testing equipment are being used by engineers, but these have a frequency limitation. Designing handheld equipment means reduction in size, which causes removal of fans, resulting in reduced frequency range.

Cabling with focus on quality. Field equipment needs to be rugged and should

Popular equipment

- R&S ZNrun for multi-port device under test
- Anritsu MTA1000 Network Master Pro for LTE network testing
- Calpods from Keysight for multi-port microwave spectrum analysis up to 110GHz
- R&S SMW200A vector signal generator for 5G testing
- Anritsu Shockline VNA for S-parameter and time-domain measurement
- VNA test cables for operations up to 40GHz from Pasternack

Operating laser in a waveguide

A pump beam is used to excite a photoconductive switch integrated in a planar waveguide, causing voltage pulses to propagate along the planar waveguide. Using Pockels electro-optic effect, the probe beam detects the electric field in the waveguide. Shape of the voltage can also be more accurately measured by varying the time delay between the two probes.